EXERCISE 6.1

1. Prove by vector method that if a line is drawn from the centre of a circle to the midpoint of a chord, then the line is perpendicular to the chord.
2. Prove by vector method that the median to the base of an isosceles triangle is perpendicular to the base.
3. Prove by vector method that an angle in a semi-circle is a right angle.
4. Prove by vector method that the diagonals of a rhombus bisect each other at right angles.
5. Using vector method, prove that if the diagonals of a parallelogram are equal, then it is a rectangle.
6. Prove by vector method that the area of the quadrilateral $ABCD$ having diagonals AC and BD is $\frac{1}{2}|\overrightarrow{AC}\times\overrightarrow{BD}|$.
7. Prove by vector method that the parallelograms on the same base and between the same parallels are equal in area.
8. If G is the centroid of a ΔABC, prove that
(area of ΔGAB) = (area of ΔGBC) = (area of ΔGCA) $=\frac{1}{3}$ (area of ΔABC).
9. Using vector method, prove that $\cos(\alpha-\beta)=\cos\alpha\cos\beta+\sin\alpha\sin\beta$.
10. Prove by vector method that $\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$,
11. A particle acted on by constant forces $8\hat{i}+2\hat{j}-6\hat{k}$ and $6\hat{i}+2\hat{j}-2\hat{k}$ is displaced from the point $(1,2,3)$ to the point $(5,4,1)$. Find the total work done by the forces.
12. Forces of magnitudes $5\sqrt{2}$ and $10\sqrt{2}$ units acting in the directions $3\hat{i}+4\hat{j}+5\hat{k}$ and $10\hat{i}+6\hat{j}-8\hat{k}$, respectively, act on a particle which is displaced from the point with position vector $4\hat{i}-3\hat{j}-2\hat{k}$ to the point with position vector $6\hat{i}+\hat{j}-3\hat{k}$ .Find the work done by the forces.
13. Find the magnitude and direction cosines of the torque of a force represented by $3\hat{i}+4\hat{j}-5\hat{k}$ about the point with position vector $2\hat{i}-3\hat{j}+4\hat{k}$ acting through a point whose position vector is $4\hat{i}+2\hat{j}-3\hat{k}$.
14. Find the torque of the resultant of the three forces represented by $-3\hat{i}+6\hat{j}-3\hat{k}$, $4\hat{i}-10\hat{j}+12\hat{k}$ and $4\hat{i}+7\hat{j}$ acting at the point with position vector $8\hat{i}-6\hat{j}-4\hat{k}$, about the point with position vector $18\hat{i}+3\hat{j}-9\hat{k}$.

6.4 Scalar triple product

Definition 6.4

For a given set of three vectors $\vec{a},\vec{b}$, and $\vec{c}$, the scalar $(\vec{a}\times\vec{b})\cdot\vec{c}$ is called a **scalar triple product** of $\vec{a},\vec{b},\vec{c}$.

Remark

$\vec{a}\cdot\vec{b}$ is a scalar and so $(\vec{a}\cdot\vec{b})\times\vec{c}$ has no meaning.

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